Farrin Marouf Sofian

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Research Interests

Training and steering generative foundation models and world models: efficient training, test-time scaling, and reward fine-tuning, with applications to images, language, and science.

Education

University of California, Irvine — Irvine, CA
Ph.D. in Computer Science — 9/2023 – 6/2028 (Expected)

- **Advisor:** Stephan Mandt
- GPA: 3.98/4.0

Koç University — Istanbul, Turkey
B.Sc. in Computer Science | Mathematics Minor — 9/2019 – 6/2023

Publications

* Denotes equal contribution.

Conference Papers

[1] **Hierarchical Variational Policies for Reward-Guided Diffusion** 2026
K. Pandey*, **FM. Sofian***, J. Groeneveld, F. Draxler, S. Mandt
Neural Information Processing Systems (**NeurIPS**) · arXiv

[2] **Control-Augmented Autoregressive Diffusion for Data Assimilation** 2026
P. Srivastava, **FM. Sofian**, F. Immorlano, K. Pandey, S. Mandt
Neural Information Processing Systems (**NeurIPS**) · arXiv

[3] **Parallel Token Prediction for Language Models** 2026
F. Draxler, J. Will, **FM. Sofian**, T. Karaletsos, S. Mandt
International Conference on Learning Representations (**ICLR**) · arXiv · Code

[4] **Variational Control for Guidance in Diffusion Models** 2025
K. Pandey*, **FM. Sofian***, F. Draxler, T. Karaletsos, S. Mandt
International Conference on Machine Learning (**ICML**) · arXiv · Code

[5] **GECTurk: Grammatical Error Correction and Detection Dataset for Turkish** 2023
A. Kara, **FM. Sofian**, A. Bond, G. G. Sahin
IJCNLP-AACL · arXiv · Code

Under Review & Preprints

[6] **Representation-Augmented Diffusions** 2026
FM. Sofian*, K. Pandey*, S. Mandt
Under review · Paper · Project

[7] **Parallel Language Models via Variational Decoding** 2026
F. Draxler*, RC. Geyer*, D. Geyfman*, **FM. Sofian***, J. Will*, et al.
Under review · arXiv · Project

[8] **Variational Test-time Optimization for Diffusion Synchronization** 2026
H. Lee*, **FM. Sofian***, K. Pandey, S. Mandt
Under review · arXiv · Project

[9] **SonicDiffusion: Audio-Driven Image Generation and Editing with Pretrained Diffusion Models** 2024
BC. Biner, **FM. Sofian**, UB. Karakaş, D. Ceylan, E. Erdem, et al.
Preprint · arXiv · Project · Code

Work & Research Experience

Mandt Lab at UC Irvine 9/2023 – Present
Graduate Researcher Irvine, California

- **Foundation Model Training**
  - Led Representation-Augmented Diffusions: **70x** faster training than a standard diffusion transformer (SiT-XL/2) and **10x** fewer sampling steps than representation alignment (REPA) at equal FID [6].
  - Led post-training of 1.3B/3B parallel LLMs on multi-node GH200 (200K GPU-hour Swiss AI grant): method design, evaluation, SFT; **2.3–5.7x** faster inference than an identically trained AR LLM [7].
- **Test-time Scaling**
  - Led a training-free optimal-control method to synchronize diffusion trajectories under a task reward, with **38%** better color consistency than the best prior synchronization method [8].
- **Reward Fine-tuning**
  - Led hierarchical variational policies that amortize diffusion guidance into a lightweight policy network, matching the best test-time optimization baseline (NDTM) with up to **7x** faster inference [1].
  - *AI for Science:* co-built control-augmented data assimilation for chaotic PDEs and real-world weather: ∼**4x** lower RMSE and **10x** faster inference than the best diffusion baselines; scales to steering a pretrained global weather model (DeepMind's GenCast) [2].

Chan Zuckerberg Initiative (CZI) 11/2024 – 9/2025
AI Research Resident Remote

- **Test-time Scaling:** led a general-purpose, training-free variational-control framework for diffusion guidance: up to **35%** better LPIPS than the best baseline, with **7x** faster runtime while extending to partially known rewards [4].
- **Advisors:** Stephan Mandt and Theofanis Karaletsos

Adobe 6/2024 – 9/2024
Machine Learning Engineer Intern San Jose, California

- Built and optimized the RAG pipeline for Adobe Acrobat's AI assistant across diverse PDF structures: **9x** faster, with up to **73 cents** lower input-token cost per question.

Awards, Honors & Grants

Swiss AI Compute Grant, 200K NVIDIA GH200 GPU hours for Parallel LMs [7] (post-training lead) 2026
Winner, Beall Butterworth Product Development Competition 2026
Selected, Chan Zuckerberg Initiative AI Residency Program 2025
UCI Dean's Fellowship 2024
Vehbi Koç Scholar for outstanding academic performance, Koç University 2019 – 2023

Mentoring & Invited Talks

Hyunsoo Lee, undergrad at Seoul National Univ.; co-developed diffusion synchronization [8] 2026
Katherine Anne Frields, undergrad at UCI (now PhD, Oxford); research onboarding 2024 – 2025
Crystal Tran & Billy Li, undergrads at UCI; led their applied data science projects 2025
Talk: *Generative Modeling with Diffusion Models: From Images to Climate Science*, Data @ UCI 2025

Teaching

CS 178 Machine Learning (2025–26) | **CS 301** Prog. Languages (2023) | **CS 132** Advanced Programming (2023)

Skills

ML & Training: PyTorch, PyTorch Lightning, Hugging Face · multi-node training (FSDP/DeepSpeed, Slurm) on GH200/H100 · diffusion & LLM post-training, RL fine-tuning
Languages: Python, C, Java, JavaScript